THE PINHOLE CAMERA MODEL

WHAT IS IT?

Action or mathematical model explaining how a 3D point from the real world is projected onto a 2D image plane using a camera.

Key Concept

- A 3D world point (X, Y, Z) is projected onto a 2D image pixel (u, v),
- This transformation depends on camera parameters.
- Intrinsic parameters include focal length and optical center.
- Extrinsic parameters represent the camera's rotation and translation in the world.

Components of the Model

- 3D World Coordinates (X, Y, Z)
- Image Coordinates (u, v)
- Intrinsic Parameters
- Extrinsic Parameters
- Camera Projection Geometry

How the Projection Works

- Light rays travel from the 3D point toward the camera.
- The pinhole acts as the projection center.
- The rays intersect the image plane.
- The intersection point becomes the image pixel (u, v).
- This forms a mathematical mapping from 3D to 2D,

The Math

The mapping from 3D world coordinates to the image plane is expressed using a projection matrix P.

The relationship is written as: $x = P X$

where X represents the 3D point in homogeneous coordinates and x represents the projected image point.

Two-View Geometry: Epipolar Geometry

WHAT IS IT?

Epipolar geometry describes the geometric relationship between two camera views observing the same scene.

When a second camera is added, depth information can be recovered from the correspondence between the two images.

Epipolar Lines

If a point appears in Image 1, its exact 3D depth is unknown.

However, the point must lie somewhere along a ray in 3D space.

In Image 2, this ray appears as a line called the **epipolar line**.

The corresponding point must lie on this line, which greatly reduces the search area for matching points.

Fundamental & Essential Matrices

- The Essential Matrix (E) represents the relationship between two calibrated cameras.
- It is based on the cameras' rotation and translation.
- The Fundamental Matrix (F) works directly with pixel coordinates.
- It combines both camera motion and camera internal parameters.
- These matrices describe the epipolar geometry between two images.

Epipolar Constraint

The relationship between corresponding points in two images is expressed using the epipolar constraint.

This constraint ensures that a point in one image must lie along the corresponding epipolar line in the other image.

The mathematical expression of this relationship is:

$$x'^T F x = 0$$

Triangulation

Once the same point is identified in both images, the two projection rays from each camera can be intersected.

The intersection of these rays gives the actual 3D coordinates of the point in space.

This process is known as triangulation and allows depth to be computed from multiple views.

Moving to Planes: Homographies

WHAT IS A HOMOGRAPHY?

A **homography** describes a transformation between two images of the same planar surface.

It provides a direct mathematical relationship between corresponding points in the two images.

Planar Scenes

If all the observed 3D points lie on a single flat plane, such as a wall, poster, or ground surface, the relationship between the two images can be described using a homography matrix.

This allows points from one image be mapped directly to the other.

Pure Camera Rotation

- A homography also applies when the camera rotates but does not change position.
- In this case, the camera orientation changes but its location remains fixed.
- Because of this, the transformation between images can still be modeled using a homography.
- This works even if the scene is fully 3D.

Homography Equation

A homography maps points from one image plane to another using a transformation matrix.

The relationship between the points is expressed mathematically as :

$$x' = Hx$$

Where **H** is the homography matrix that transforms the coordinates of a point from the first image into the second image.

Applications

- Image stitching to create panoramas.
- Perspective correction in photographs.
- Mapping planar surfaces between images.
- Augmented reality and computer vision tasks.

Scaling Up: Structure from Motion (SfM)

WHAT IS SfM?

Structure from Motion (SfM) is a computer vision technique that reconstructs a 3D scene from multiple overlapping 2D images.

It simultaneously estimates the 3D structure of the scene and the positions of the cameras that captured the images.

Multiple View Geometry

After understanding the relationship between two views, the same principles can be extended to many images.

By analyzing correspondences between points across several images, the system can recover both the camera motion and the scene structure.

This allows large environments to be reconstructed in 3D.

What SfM Estimates

- 3D coordinates of scene points
- Positions and orientations of cameras
- Relationships between multiple views
- Depth information from overlapping images
- A consistent 3D reconstruction of the scene

Bundle Adjustment

Bundle adjustment is a crucial optimization step in Structure from Motion.

It refines the estimated camera positions and the 3D coordinates of scene points.

The goal to minimize the reprojection error, which measures the difference between where a projected 3D point should appear in the image and where it was actually detected.

Why It Matters

- Used for 3D reconstruction from photos
- Important in photogrammetry and mapping
- Used in robotics and autonomous navigation
- Enables large-scale scene modeling from images.